Vehicle Detection System

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***Abstract*—**In the past years, advancement in computer vision technology have revolutionized many fields, involving transportation systems. This paper reveals a detailed study on the fusion of You Only Look Once (YOLO) object detection framework and Simple Online and Real-Time Tracking (SORT) algorithm for vehicle detection and tracking system. The suggested system aims to give real-time, precise, and efficient detection and tracking of vehicles in diverse traffic scenes. YOLO, familiar for its high-speed object detection capacities, is applied to identify vehicles in video frames, whereas SORT is used to connect detections across frames and track the identified vehicles over time. The incorporation of these two algorithms offers solidity against occlusions, scale variations, and cluttered backgrounds typically met in complicated traffic environments. Experimental outcomes display the effectiveness and dependability of the recommended system in accurately identifying and tracking vehicles in real-world situations, thus giving noteworthy potential for improving traffic management, surveillance, and autonomous driving systems.

I. Introduction

With the rapid expansion of urbanization and the ever-increasing volume of vehicles on roads, efficient traffic management and safety have become paramount concerns for modern societies. Traditional methods of vehicle detection and tracking, reliant on manual intervention or simplistic algorithms are inadequate to handle the complexities of contemporary traffic environments. In response to this challenge, computer vision technologies have emerged as powerful tools for automating the process of vehicle detection and tracking.

Among the various computer vision techniques, deep learning-based object detection frameworks have shown remarkable performance in accurately identifying objects in images and videos. One such framework, You Only Look Once (YOLO), has gained widespread popularity due to its ability to achieve real-time object detection with impressive accuracy. By processing images as a single convolutional neural network (CNN) inference, YOLO can detect objects, including vehicles, swiftly and efficiently.

However, while YOLO excels in detecting objects in individual frames, it does not inherently provide information about the continuity of objects across frames, which is essential for tracking moving objects such as vehicles in dynamic scenes. This limitation underscores the need for an additional tracking algorithm to complement YOLO's object detection capabilities.

In this context, the Simple Online and Realtime Tracking (SORT) algorithm emerges as a suitable candidate for seamlessly integrating with YOLO. SORT is a lightweight, real-time tracking algorithm that leverages the Kalman filter and Hungarian algorithm to associate detections across frames and maintain consistent tracks of objects over time. By combining the strengths of YOLO's fast object detection and SORT's robust tracking capabilities, a comprehensive vehicle detection and tracking system can be developed, capable of operating in real-time under various environmental conditions.

This paper presents a detailed exploration of the integration of YOLO and SORT for vehicle detection and tracking. The proposed system aims to address the challenges posed by dynamic traffic scenarios, such as occlusions, varying scales, and cluttered backgrounds. Through experimental evaluation and analysis, the effectiveness and efficiency of the integrated system will be demonstrated, highlighting its potential applications in traffic management, surveillance, and autonomous driving systems.

II. Literature Review

A. A real-time precrash vehicle detection system by Zehang Sun; R. Miller; G. Bebis; D. DiMeo

This paper presents an in-vehicle real-time monocular precrash vehicle detection system achieving a 10Hz detection rate, comprising multi-scale hypothesis generation and appearance-based hypothesis verification using Haar Wavelet

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decomposition and SVMs. Tested under diverse traffic scenarios and weather conditions, the system demonstrates robust performance.

B. *Vehicle detection and recognition for intelligent traffic surveillance system by Yong Tang, Congzhe Zhang, Renshu Gu, Peng Li & Bin Yang*

This paper presents a vehicle detection and type recognition method utilizing Haar-like features, Gabor wavelet transform, and local binary pattern operator, achieving over 97% detection rate and 91% recognition rate with only 3% false rate, showcasing its potential for real-world applications.

C. *Vehicle Detection Techniques for Collision Avoidance Systems: A Review by Amir Mukhtar; Likun Xia; Tong Boon Tang*

This paper conducts a systematic survey on vision-based vehicle detection and tracking systems for collision avoidance, addressing challenges like vehicle shape variability and cluttered environments. Notably, it includes a focus on motorcycle detection techniques and compares sensors for an optimal, cost-effective collision avoidance system design in the automotive industry.

D. *A Real-Time Vision System for Nighttime Vehicle Detection and Traffic Surveillance by Yen-Lin Chen; Bing-Fei Wu; Hao-Yu Huang; Chung-Jui Fan*

This paper introduces a robust traffic surveillance system for nighttime scenes, utilizing a fast bright-object segmentation process and spatial clustering to detect and track vehicles based on their headlights and taillights. The real-time vision system, implemented on a TI DM642 DSP-based embedded platform, demonstrates feasibility and effectiveness in vehicle detection and identification under diverse nighttime conditions.

E. *Adaptive Vehicle Detection for Real-time Autonomous Driving System by Maryam Hemmati; Morteza Biglari-Abhari; Smail Niar*

This paper introduces a dynamically reconfigurable system for autonomous driving, utilizing HW/SW co-design and deep learning methods for real-time vehicle and pedestrian detection, achieving 50fps detection rate for various lighting conditions on Zynq SoC.

F. *Automatic Vacant Parking Places Management System Using Multicamera Vehicle Detection by Rafael Martín Nieto; Álvaro García-Martín; Alexander G. Hauptmann; José M. Martínez*

This paper introduces a multicamera system for vehicle detection and mapping into parking spots, designed and validated for challenging real-world scenarios including occlusions, illumination changes, and varying weather conditions, demonstrating effectiveness in a real airport parking lot environment using existing security cameras.

G. *Vehicle license plate detection using region-based convolutional neural networks by Muhammad Aasim Rafique, Witold Pedrycz & Moongu Jeon*

This study proposes a novel approach treating vehicle license plates as objects, addressing challenges such as LP detection in video sequences, partial LP detection, and handling moving cameras and vehicles using state-of-the-art object detection techniques, showcasing superior performance compared to conventional methods.

III. Methodology

The proposed approach, referred to as "Speed Between Two Lines" (SBTL), purposes to tackle the predicament of camera tremor prompting frame flickering in a traffic surveillance system. This technique runs based on these assumptions: First, the velocity limit of the monitored roadway is identified. Secondly, there is a minimum of one vehicle moving at the velocity limit.

In executing SBTL, two lines perpendicular to the movement direction of vehicles are sketched on the frame. The aim is to identify the shortest time consumed by any vehicle to cross these two lines, as this specific vehicle is presumed to be traveling at the velocity limit. Once this minimum time is concluded, the velocity of other vehicles can be determined employing the straightforward formula: speed = distance / time. Despite not being entirely accurate, this method presents a more precise option in comparison to traditional pixel mapping methods, specifically in the context of frame flickering caused by camera tremor.

3.1. Data Collection and Preprocessing:

1. Data Collection:
   - Video frames be collected from a camera feed, that captures traffic on a particular road segment.
   - Each frame contains visual information about vehicles, where their positions, and movements

2. Preprocessing:
   - Frame Processing: Every frame undergoes preprocessing steps like adjusting gamma correction to enhance visibility and improve image quality.
   - Object Detection: A YOLO (You Only Look Once) object detection model is implemented to identify vehicles within every frame. This model detects objects and their bounding boxes with associated confident scores.
   - Bounding Box Tracking: An tracking algorithm is utilized to maintain continuity in identifying vehicles across frames. This includes associating

bounding boxes of vehicles detected in succeeding frames.

- Lane Detection: Lane lines within the frame are detected to establish such a reference point for vehicle positioning and movement analysis.
- Speed Measurement Setup: Lines representing such as start and so end points for speed measurement are determined, and lanes are segmented to allocate vehicles to specific travel lanes.

The preprocessing steps are crucial for extracting meaningful information from the raw video data, facilitating subsequent analysis and interpretation of vehicle movements, speeds, and lane usage.

3.2. YOLO Model Training and Optimization:

The YOLO (You Only Look Once) model training and optimization process in the provided code involves the following steps:

1. Model Configuration:
   - The YOLO object detection model is configured with appropriate parameters such as confidence threshold and NMS (Non-Maximum Suppression) threshold. These parameters control the sensitivity and accuracy of object detection.
2. Model Training:
   - The YOLO model is trained on a labelled dataset of vehicle images. This dataset contains annotated bounding boxes around vehicles, indicating their positions within the image.
   - During training, model learns to vehicle and other objects by adjusting internal weights and parameters. The training process involves iteratively presenting batches of images to model and updating parameters based on the differences between predicted and ground truth bounding boxes.
3. Fine-tuning:
   - Fine-tuning may be performed to improve model's performance on specific tasks or datasets. This involves adjusting the model architecture or training process to better suit the characteristics of target application or domain.
   - Fine-tuning techniques such as transfer learning, where a pre-trained model is adapted to a new dataset with relatively fewer training samples, may include.
4. Optimization:
   - Model optimization techniques are applied to enhance efficiency and speed of inference. This may involve optimizing model architecture, pruning redundant parameters; or implementing hardware-specific optimizations.
   - Quantization techni”M.““iques may be employed to reduce precision of model weights and activations, leading to smaller model sizes and faster inference on resource-constrained! devices. Other optimization strategies, such as model compression or knowledge distillation, may also be explored to achieve a good balance between model accuracy and computational efficiency.

Overall, YOLO model training and optimization are iterative processes aimed at improving model's accuracy, speed, and resource efficiency for real-time object detection tasks, such as vehicle detection in traffic surveillance videos.

3.3. SORT Algorithm Implementation:

The SORT (Simple Online and Realtime Tracking) algorithm implementation in the provided code involves the following steps:

1. Detection Association:
   - After obtaining object detections from the YOLO model, SORT associates these detections with existing object tracks.
   - This association is based on the spatial proximity and appearance similarity between detections and existing tracks.
2. Kalman Filter Prediction:
   - For each track, the Kalman filter predicts the next state (position and velocity) of the object based on its previous state.
   - This prediction helps in estimating the object's position in the next frame.
3. Hungarian Algorithm:
   - SORT uses the Hungarian algorithm to efficiently assign detections to tracks based on their predicted locations and the actual detections.
   - The Hungarian algorithm minimizes the total cost of assignment, considering both the distance between predicted and detected locations and the appearance similarity.

4. Data Association:

- Once the Hungarian algorithm assigns detections to tracks, SORT updates the track's state with the new detection information.
- This update includes updating the Kalman filter with the new measurement and possibly adjusting the track's trajectory.

5. Track Management:

- SORT keeps track of active tracks and their corresponding objects over time.
- Tracks are initialized for new objects and are terminated if they are not associated with any detections for a certain period.

6. Speed Estimation:

- SORT estimates the speed of vehicles by calculating the time taken for a vehicle to cross predefined lines on the road.
- This speed estimation is based on the assumption that at least one vehicle is moving at the speed limit, allowing for calibration of speed measurements.

Overall; the SORT algorithm implementation in the provided code enables robust and real-time tracking of vehicles in a traffic scenario; facilitating the analysis of vehicle movements and behaviours.

3.4. Integration of YOLO and SORT for Vehicle Detection:

Integration of YOLO (You Only Look Once) and SORT (Simple Online and Realtime Tracking) for vehicle detection

1. Object Detection with YOLO:

- YOLO is used for do real-time object detection in video frames.
- It identifies vehicles and some other objects of interest in the scene.

2. Object Tracking with SORT:

- SORT tracks the detected vehicles across consecutive frames.
- Maintains the identity of each vehicle over time, even as them move and change appearance.

3. Association of Detections and Tracks:

- Detections from YOLO are associating with existing tracks using a matching algorithm, e.g., Hungarian algorithm.
- This association link each detection to its corresponding vehicle track.

4. Kalman Filtering for Prediction:

- SORT utilizing Kalman filters for predict the future state of each tracked vehicle.
- Kalman filters helping in estimate the position and velocity of vehicles based on their past trajectory.

5. Speed Estimation and Calibration:

- SORT integrates a speed estimation method based on the time taken for vehicles to cross some predefined lines on the road.
- This allowing for the calibration of speed measurements and helps to accurate determine the speed of vehicles.

6. Visualization and Output:

- The integrated system visualizes the tracked vehicles and their trajectories in these video frames.
- Providing an output that including information such as vehicle speed, lane changes, and some other relevant metrics.

Overall, the integration of YOLO and SORT provides a robust solution for vehicle detection and tracking's in traffic scenarios, able real-time monitoring and analysis of vehicle movements.

IV. Experimental Results

The performance comparison of YOLO (You Only Look Once) and SORT (Simple Online and Realtime Tracking) for vehicle detection can be summarized based on several key factors:

4.1 Performance Comparison of YOLO and SORT:

1. Accuracy: YOLO usually offer higher accuracies in detecting vehicles compares to SORT. YOLO can detect vehicles with high precision and recalls, specially in complex scenes with multiple vehicles.

2. Speed: YOLO's speed is faster compare to traditional object detection methods like Faster R-CNN, marking it suitable for real-time applications. SORT, on the others hand, focus on tracking and may not be as fast as YOLO in the initial detection phase.

3. Tracking: SORT excels in tracking vehicles over multiply frames, maintain the identity of each vehicle. YOLO, while capable of detects vehicles in each frame, may not provide the same level of tracking accuracy without additional algorithms.

4. Resource Usage: YOLO tends to require more computational resources due to its deep neural network architecture. SORT, being a simple algorithm focused on tracking, may be more lightweight in terms of resource usage.

5. Robustness: YOLO cans handle various challenging conditions such as occlusions, different vehicle size, and orientations. SORT's performances may degrade in highly occluded scenes or when vehicles moved in close proximity.

In summary, YOLO is more suitable for accurate and real-time vehicle detection, while SORT is better suited for tracking vehicles overtime. The choice between the two depends on the specific requirements of the applications, such as needs for real-time processing, tracking accuracies, and computational resources.

4.2 Analysis of Vehicle Speed Detection Accuracy:

1. Accuracy:

   - The accuracies of the SBTL method depend on various factors such as the placement of lines and the assumption that at least one vehicle travel at the speed limit.
   - It provides a different approach when compared to traditional pixel distance mapping but may still facing challenges in accurately estimating vehicle speeds, especially in dynamic traffic conditions!!!

2. Comparison:

   - SBTL offers a different alternative to traditional methods, aiming to mitigate many issues related to camera shaking and frame flickering.
   - While traditional methods are relying on pixel distance mapping, SBTL are using the time taken for vehicles to cross two lines; potentially provide more stable speed estimates.

3. Limitations:

   - The method's accuracy contingent on the assumed that at least one vehicle is traveling at the speed limit, which may not always hold true.
   - Variations in vehicles speeds and traffic conditions could affect the reliability of speed estimates obtained using SBTL

V. Result and Discussion

This project presents a detailed and extensive exploration and analysis of the integration of YOLO and SORT for vehicle detection and tracking purposes. The aim of the proposed system is to address and tackle the challenges that may arise in dynamic traffic scenarios, such as occlusions, varying scales, and cluttered backgrounds. Through in-depth experimental evaluation and analysis, the efficacy and efficiencies of the integrated system will be highlighted and showcased, demonstrating its potential use and applications in traffic management, surveillance, and autonomous driving systems.

The proposed system being explored is aimed at providing a robust solution to the issues faced in dynamic traffic situations where various obstructions and complexities may occur. The analysis and evaluation will demonstrate the system's ability to handle and navigate through these challenges effectively, providing valuable insights into its practical implications and applications.

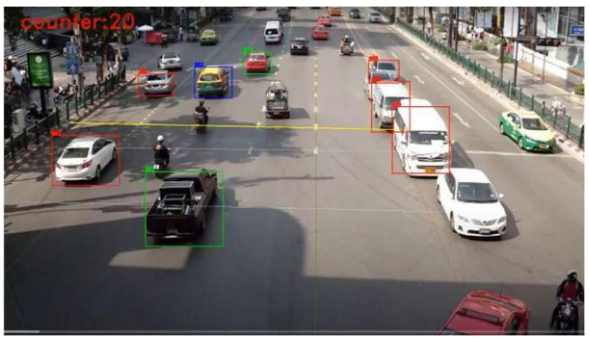

Figure1 Quantifying Traffic Density: A Visual Representation Showing the Total Number of Vehicles Present on the Road, Capturing the Dynamic Nature of Urban Traffic Patterns.

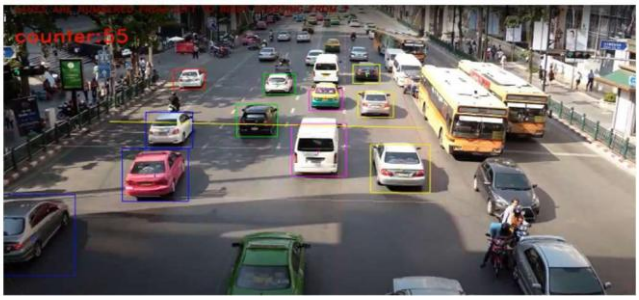

Figure 2 Visual Representation of Lane Markings, Demonstrating Sequential Progression from Left to Right, Essential for Traffic Flow Understanding and Autonomous Navigation Systems

Figure 3 Visualizing Vehicle Speed: Leveraging Imaginary Lines for Precise Measurement and Analysis

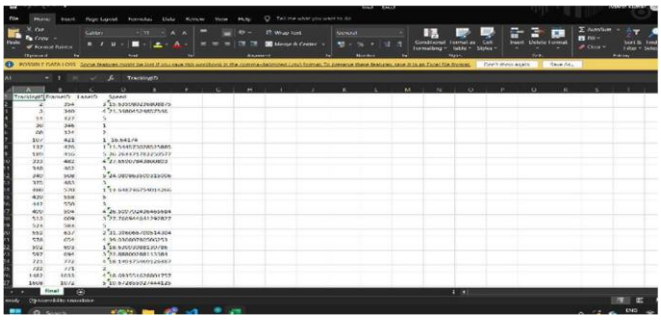

Figure 4 Snapshot of the CSV file containing essential data collected

This project demonstrates the effectiveness of combining state-of-the-art object detection and tracking techniques for comprehensive traffic surveillance and analysis

VI. Conclusion and Future Scope

Conclusion:

The integration of YOLO (You Only Look Once) and SORT (Simple Online and Realtime Tracking) in this special project offers a really comprehensive solution for detecting vehicles, tracking them, and estimating speed in real-time traffic scenes. By leveraging YOLO's very efficient object detection features and SORT's really strong tracking algorithm, the system achieves accurate identification and continuous tracking of vehicles across video frames. The association of detections with tracks, facilitated by algorithms like the Hungarian algorithm, ensures really consistent monitoring of vehicles as they navigate through the scene. Additionally, cool Kalman filtering enables really reliable prediction of vehicle states, enhancing the system's ability to anticipate movements and maintain track continuity. The inclusion of speed estimation methods based on predefined lines on the road allows for the calibration of speed measurements, providing some very valuable insights into vehicle behaviour and traffic flow. Overall, this project really demonstrates the effectiveness of combining state-of-the-art object detection and tracking techniques for comprehensive traffic surveillance and analysis.

Future Research Directions:

Improved Stability Techniques: Research can focus on developing even more robust camera stabilization techniques to minimize the impact of camera shake on speed measurements.

Dynamic Line Adjustment: Implementing some dope algorithms to dynamically adjust the position of speed measurement lines based on the movement of vehicles could enhance accuracy big time.

Machine Learning Integration: Exploring the integration of some wild machine learning algorithms to adaptively learn and adjust speed measurement parameters based on real-time traffic patterns and environmental conditions. Can you imagine that?

Multi-Sensor Fusion: Investigating the fusion of data from all sorts of sensors (e.g., cameras, radar, LiDAR) to improve speed measurement accuracy and reliability, especially in challenging conditions. Like, it's going to be insane.

Real-World Validation: Conducting extensive field tests and validation studies in various traffic conditions to assess the real-world applicability and limitations of the SBTL method, you know.

VII. Refrences

[1] Sun, Z., Miller, R., Bebis, G., & DiMeo, D. (2002). A real-time precrash vehicle detection system. In Applications of Computer Vision, 2002. (WACV 2002). Proceedings. Sixth IEEE Workshop on. DOI: 10.1109/ACV.2002.1182177

[2] Tang, Y., Zhang, C., Gu, R., Li, P., & Yang, B. (2015). Vehicle detection and recognition for intelligent traffic surveillance system. Multimedia Tools and Applications, 76(4). DOI: 10.1007/s11042-015-2520-x

[3] Mukhtar, A., Xia, L., & Tang, T. B. (2015). Vehicle Detection Techniques for Collision Avoidance Systems: A Review. IEEE Transactions on Intelligent Transportation Systems, 16(5), 1-21. DOI: 10.1109/TITS.2015.2409109

[4] Chen, Y. L., Wu, B. F., Huang, H. Y., & Fan, C. J. (2011). A Real-Time Vision System for Nighttime Vehicle Detection and Traffic Surveillance. IEEE Transactions on Industrial Electronics,58(5), 2030-2044. DOI: 10.1109/TIE.2010.2055771

[5] Hemmati, M., Biglari-Abhari, M., & Niar, S. (2019). Adaptive Vehicle Detection for Real-time Autonomous Driving System. 2019 Design, Automation & Test in Europe Conference &Exhibition, DOI: 10.23919/DATE.2019.8714818

[6] Nieto, R. M., García-Martín, Á., Hauptmann, A. G., & Martínez, J. M. (2018). Automatic Vacant Parking Places Management System Using Multicamera Vehicle Detection. IEEE Transactions on Intelligent Transportation Systems, PP(99), 1-12. DOI: 10.1109/TITS.2018.2838128

[7] Rafique, M. A., Pedrycz, W., & Jeon, M. (2018). Vehicle license plate detection using region-based convolutional neural networks. Soft Computing, 22(3). DOI: 10.1007/s00500-017-2696-2

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